The University of Texas at Austin
Optimization

Homework 8 Solutions

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1. **Subgradients**. Show the following for sub-gradients

   (a) If $g(x) = f(Ax+b)$, then $\partial g(x) \supseteq A^T \partial f(Ax+b)$.

   Solution:

   $A^T \partial f(Ax+b) \subseteq \partial g(x)$:

$$\begin{aligned} f(Ay+b) &\geq f(Ax+b) + \langle \partial f(Ax+b), A(y-x) \rangle \\ &= f(Ax+b) + \langle A^T \partial f(Ax+b), y-x \rangle \\ &\iff \\ g(y) &\geq g(x) + \langle A^T \partial f(Ax+b), y-x \rangle \end{aligned}$$

   Thus, $A^T \partial f(Ax+b) \subseteq \partial g(x)$.

   $\partial g(x) \subseteq A^T \partial f(Ax+b)$: *Let* $z \in \partial g(x)$. *Consider the following optimization problem:*

$$\begin{aligned} \min_{v,y} \quad & f(v) - g(x) - \langle z, v-x \rangle \\ s.t. \quad & v = Ay + b \end{aligned}$$

   Consider the Lagrangian of this problem

$$L(v,y,\lambda) = f(v) - g(x) - \langle z, y-x \rangle + \lambda^T (Ay+b-v).$$

   We know $y = x, v = Ax+b$ *is the solution of primal problem and strong duality holds since slater's condition holds. Let* λ^* *be the dual optimal solution. Since* $z \in \partial g(x)$, *the optimal value of primal problem is 0. Consequently,* $L(v,y,\lambda^*) \geq 0$ *for all* v, y. *It implies* $z = A^T \lambda^*$ *(also can be derived from KKT condition). Thus,*

$$\begin{aligned} L(v,y,\lambda^*) &= f(v) - g(x) - \langle z, y-x \rangle + \lambda^*(Ay+b-v) \\ &= f(v) - g(x) + \langle A^T \lambda^*, x \rangle + \langle \lambda^*, b-v \rangle \\ &= f(v) - f(Ax+b) + \langle \lambda^*, Ax+b-v \rangle \geq 0, \end{aligned}$$

   for all v. *It implies* $\lambda^* \in \partial f(Ax+b)$. *Therefore,* $z \in \partial A^T f(Ax+b)$, *proving* $\partial g(x) \subseteq A^T \partial f(Ax+b)$.

   (b) If $f(x) = \max_{1 \leq i \leq m} f_i(x)$, then $\partial f(x) \supseteq \operatorname{conv}\left(\bigcup_{i: f_i(x) = f(x)} \partial f_i(x)\right)$.

   Solution:

$\mathbf{conv}\left(\bigcup_{i:f_i(x)=f(x)} \partial f_i(x)\right) \subseteq \partial f(x)$ *: For any $z \in conv\left(\bigcup_{i:f_i(x)=f(x)} \partial f_i(x)\right)$, it can be represented as convex combination of z_i where each $z_i \in \partial f_i(x)$. Then, for each $i : f_i(x) = f(x)$,*

$$f_i(y) \geq f_i(x) + \langle z_i, y - x\rangle,$$

and the convex combination will imply

$$\sum_i c_i f_i(y) \geq \sum_i c_i f_i(x) + \sum_i c_i \langle z_i, y - x\rangle,$$

therefore,

$$\begin{aligned} f(y) \geq \sum_i c_i f_i(y) &\geq \sum_i c_i f_i(x) + \langle \sum_i c_i z_i, y - x\rangle \\ &= f(x) + \langle z, y - x\rangle, \end{aligned}$$

for all y, which implies $z \in \partial f(x)$.

$\partial f(x) \subseteq \mathbf{conv}\left(\bigcup_{i:f_i(x)=f(x)} \partial f_i(x)\right)$ *: Consider the following optimization problem:*

$$\begin{aligned} \min_{y,\gamma} \quad & \gamma - f(x) - \langle z, y - x\rangle \\ s.t. \quad & f_i(y) \leq \gamma \qquad \forall i, \end{aligned}$$

where $z \in \partial f(x)$. Using KKT condition with associated optimal dual variables $\lambda_i^* \geq 0$, we have

$$L(y, \gamma, \lambda_i^*) = \gamma - f(x) - \langle z, y - x\rangle + \sum_i \lambda_i^*(f_i(y) - \gamma)$$

and thus,

$$\sum_i \lambda_i^* = 1, \quad \lambda_i^* \geq 0.$$

Also, we know that one optimal solution is $y^* = x$ since $z \in \partial f(x)$. Thus, taking gradient of Lagrangian with respect to y at x gives,

$$\sum_i \lambda_i^* \partial f_i(y) = z,$$

which implies z is convex combination of subgradient of each function.
Finally, we have that

$$L(x, f(x), \lambda_i^*) = -f(x) + \sum_i \lambda_i^* f_i(x) = 0,$$

as the optimal value is 0. It implies only λ_i^* where $f_i(x) = f(x)$ can be non-zero. Thus, we get the result.

Thus, as we have shown here, the containments above all equalities.

2. **More Subgradients**. Show that for any convex function f, and for any points x, y, and $u \in \partial f(x)$, $v \in \partial f(y)$,

$$\langle u - v, x - y \rangle \geq 0.$$

This shows that the subdifferential is a monotone operator for any convex function. We will not use this specific property in this class, but it is heavily used for more advanced aspects of convex optimization and convex analysis.

Solution: *By the first order condition for convex functions, for any points x, y, and $u \in \partial f(x)$, $v \in \partial f(y)$,*

$$f(y) \geq f(x) + \langle u, y - x \rangle$$
$$f(x) \geq f(y) + \langle v, x - y \rangle.$$

Summing up two inequalities, we get $\langle u - v, x - y \rangle \geq 0$ as desired. In fact, the converse is also true: monotonicity implies convexity.

3. **Coordinate Descent**

(a) Give an example that shows that coordinate descent may not find the optimum of a convex function. That is, provide a simple function f and a point x such that coordinate descent starting from x will *not* get to the global minimum of f.

Solution: *One of the simplest example is $f(x_1, x_2) = \max\{x_1, x_2\}$. If coordinate descent starts from $(1, 1)$, it will not get to the global minimum $(0, 0)$. The algorithm stops at $(1, 1)$ because each of x_1 and x_2 is at a local minimum.*

(b) Let $f(x, y) = x^2 + y^2 + 3xy$, where x, y are scalars. Note that f is not convex. Would coordinate descent with exact line search always converge to a stationary point ?

Solution: *Consider the partial derivatives*

$$\frac{\partial f}{\partial x} = 2x + 3y, \quad \frac{\partial f}{\partial y} = 2y + 3x.$$

We see that $\partial f / \partial x = 0$ for $x = -3y/2$, and $\partial f / \partial y = 0$ for $y = -3x/2$. Since the coordinate descent with exact line search finds the point at which the partial derivative is zero, the update is given by

$$x_{k+1} = -3y_k/2, \; y_{k+1} = -3x_{k+1}/2.$$

This shows that each coordinate will increase geometrically with a factor $9/4$. This algorithm does not converge to a stationary point.

4. **Frank Wolfe and PGD**. In the lectures, we showed an example where Frank-Wolge and projected gradient descent (PGD) behave very differently. Replicate that here, and show your plots of the trajectory obtained.

$$\begin{aligned} \min: \quad & 100x_1^2 + x_2^2 + (x_3 - 20)^2 \\ \text{s.t.}: \quad & x_1 + x_2 + x_3/20 = 1 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Solution: *The key intuition behind this example, and why we expect to have PGD and F-W have very different trajectories (especially for small step size) is as explained in the lecture. Another important reason you would use Frank Wolfe is the setting where solving the LP is much less expensive than computing the projection.*